PassFrame tuition

Coordinates & Transformations

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. -4

2. 6

3. (3, 3)

4. (4, 3)

5. A — (-2, 2)

6. The third quadrant (both x and y are negative).

7. The fourth quadrant (x is positive, y is negative).

8. (-2, -3)

9. (-4, 6)

10. A, B and D (P(-2, 4), Q(2, -4), S(2, 4)) — R is the only point where both coordinates are negative (the third quadrant)

11. (-3, -2)

12. A — (2, 2)

13. (-6, -3)

14. (-2, 3)

15. A — (0, -7)

Standard

1. (-4, -5)

2. (-6, -3)

3. (5, 5)

4. A — (-6, 2)

5. Reflect in x-axis: (5, -4). Reflect in y-axis: (-5, -4).

6. (8, -4)

7. 8 units (from x = -4 to x = 4, along y = 3).

8. (-3, -2)

9. A — (-7, -5)

10. (-9, 6)

11. A, B and C (Its width is 8 units, Its height is 6 units, The vertex (5, 4) is diagonally opposite (-3, -2))

12. (9, 9)

13. Reflect: (-2, -6). Translate 3 down: (-2, -9).

14. A — (3, -3)

15. (-7, -7)

Standard / Hard

1. (5, -2)

2. (3, 4)

3. (3, -6)

4. A — (-5, -2)

5. (y, -x) = (-6, 0).

6. (8, -5)

7. Side length 4 units; centre at (0, 0), the origin.

8. (5, -3)

9. (-7, -9)

10. B, C and D (B becomes (1, -4), C becomes (-3, -4), D becomes (-3, 0))

11. (-8, 6)

12. Reflect: (-4, -6). Rotate: (y, -x) = (-6, 4).

13. A — 22 units (width 6, height 5, perimeter = 2 × (6 + 5) = 22)

14. (1, -9)

15. 6 right and 6 up.

Homework — Answers

Easy / Standard

1. -6

2. 2

3. (3, 5)

4. (2, -3)

5. A — (3, -4)

6. The second quadrant (x is negative, y is positive).

7. The fourth quadrant (x is positive, y is negative).

8. (-5, -3)

9. (-4, 3)

10. B, C and D (Q(3, -5), R(-3, -5), S(3, 5)) — P is the only point with a negative x-coordinate and a positive y-coordinate (the second quadrant)

11. (-4, -4)

12. A — (2, 5)

13. (-8, -5)

14. (-4, 4)

15. A — (-8, 0)

Standard

1. (-5, -6)

2. (-7, -4)

3. (4, 4)

4. A — (-8, 3)

5. Reflect in x-axis: (6, -5). Reflect in y-axis: (-6, -5).

6. (9, -5)

7. 10 units (from x = -5 to x = 5, along y = 2).

8. (-3, -2)

9. A — (-8, -6)

10. (-10, 7)

11. A, B and C (Its width is 10 units, Its height is 8 units, The vertex (6, 5) is diagonally opposite (-4, -3))

12. (9, 9)

13. Reflect: (-3, -7). Translate 4 down: (-3, -11).

14. A — (3, -3)

15. (-8, -8)

Standard / Advanced

1. Rotate: (x, y) → (y, -x) = (-6, -3). Translate 2 left: (-8, -3).

2. It needs to travel 5 right to reach x = 0 (its x-coordinate is -5).

3. Width = 6 - (-4) = 10, height = 3 - (-2) = 5. Perimeter = 2 × (10 + 5) = 30 units.

4. Reflect in x-axis: (-2, -2) → (-2, 2). Sum = -2 + 2 = 0.

5. 8 left: 8 - 8 = 0, giving (0, -3), which lies on the y-axis.

6. (7, 4)

7. Side length 6 units; centre at (0, 0), the origin.

8. (2, -6)

9. (9, 8)

10. B, C and D (B becomes (2, -5), C becomes (-4, -5), D becomes (-4, 0))

11. (-8, 6)

12. Reflect: (-5, -8). Rotate: (y, -x) = (-8, 5).

13. A — 30 units (width 8, height 7, perimeter = 2 × (8 + 7) = 30)

14. (-2, 11)

15. 8 right and 8 up.